Savitribai Phule Pune University
T.Y.BCA (SEM-V) NEP
Practical Examination OCT / NOV
CA-305-MJP: Fundamentals of Data Science

Duration: 3 Hours Maximum Marks: 35

Q1) **A** Create a DataFrame from a dictionary containing employee name, department, designation, and salary for at least 8 employees. Display only the first 5 records using a suitable Pandas function. [10 marks]

Q2) **B** Write a program to create Dataset Name: StudentData.csv [20 marks]

Save the following data in Excel and store it as a .CSV file.

Name	Address	Age	Class
Smaran	Pune	20	BE
Sachin	Mumbai	23	BSc(CS)
Samarpan	Delhi	19	BVoC
Spandan	Chennai	21	BCom
Shravan	Madurai	22	BSc

Import Dataset

The dataset contains two categorical columns:

- Name
- Class

Perform:

a) One-Hot Encoding on **Nam**e column

b) Label Encoding on **Class** column

OR

Q2) **B** Write a program to generate a random array of 25 integers and display them using a line chart, scatter plot, histogram and box plot. Apply appropriate color, labels and styling options [20 marks]

Q3) Viva [05 marks]

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Q1) **A** Write a Python program using Pandas to calculate the sum, average, median, mode, minimum, maximum, and standard deviation of the following marks: [10 marks]

Marks = [72, 85, 91, 68, 85, 79, 94]

Q2) **B** Write a program to create two lists, one representing student names and the other representing percentage. Display the data in a pie chart and bar chart. [20 marks]

OR

Q2) **B** Write a program to create a DataFrame containing 2D coordinates of multiple data points. Write a Python program to select any two points and compute the Euclidean distance between them using numpy.linalg.norm(), showing both the difference vector and the final distance. [20 marks]

Q3) Viva [05 marks]

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Q1) **A** Create a DataFrame from a dictionary containing product name, brand, category and price for at least 8 products. Display only the first 5 records using a suitable Pandas function. [10 marks]

Q2) **B** Write a program to create a DataFrame containing the salaries (in thousands) of 15 employees in a company. Write a Python program to calculate the quartiles (Q1, Q2, Q3) and selected percentiles (25th, 50th, 75th, 90th) and detect outliers using both the IQR and Z-score methods. [20 marks]

OR

Q2) **B** Write a program to create Dataset Name: StudentData.csv [20 marks]

Save the following data in Excel and store it as a .CSV file.

Name	Address	Age	Class
Shrikrishna	Trivendram	20	BE
Shridhar	Madurai		BSc(CA)
Mohan	Suchindram	19	BVoC
Govind	Rameswaram	21	BCom
Vasudev	Kanyakumari	22	BSc

Import Dataset

Replace missing values of:

- Age column → with mean age

Q3) Viva [05 marks]

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Q1) **A** Create a dictionary containing a list of laptop models and their corresponding prices. Use the keys Laptop_Model and Cost, convert the dictionary into a Pandas DataFrame and display it. [10 marks]

Q2) **B** Write a program to create two lists, one representing student names and the other representing percentage. Display the data in a pie chart and bar chart. [20 marks]

OR

Q2) **B** Write a program to create a DataFrame containing the monthly sales (in Lakh) of a Supply Chain Management System for all months. Write a Python program to calculate range, variance and standard deviation of sales data. [20 marks]

Q3) Viva [05 marks]

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Q1) **A** Create a DataFrame from a dictionary containing student name, age, course and marks for at least 8 students. Display only the first 5 records using a suitable Pandas function. [10 marks]

Q2) **B** 1. To create a dataframe containing columns emp_name, age and salary. Add 10 rows to the dataframe. [20 marks]

2. Apply shape (), describe () on given frame

3. To Add 5 rows with duplicate values and missing values. Add a column 'Department' with empty values. Display the data.

OR

Q2) **B** Write a program to create Dataset Name: StudentData.csv [20 marks]

Save the following data in Excel and store it as a .CSV file.

Name	Address	Age	Class
Smaran	Pune	20	ME
Sachin	Mumbai	23	MSc (CA)
Samarpan	Delhi	19	MVoC
Spandan	Chennai	21	MCom
Shaman	Madurai	22	MSc

Import Dataset and Perform Basic Exploration

a) Display statistical description of dataset

b) Display shape of dataset

c) Display first two rows

Q3) Viva [05 marks]

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Q1) **A** Create a file named staff.csv containing staff_id, employee_name, department, and monthly_salary for at least 5 employees. Read the CSV file using Pandas and display it as a DataFrame. [10 marks]

Q2) **B** Write a program to create a NumPy array of the ages of 15 employees and write a Python program to compute and display the histogram using custom bins, showing the frequency of ages in each bin and plot the histogram using matplotlib with labelled axes. [20 marks]

OR

Q2) **B** Write a program to create a student DataFrame which contains Roll No, Name and Percentage. Write a Python program to calculate mean, median and mode of given data. [20 marks]

Q3) Viva [05 marks]

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Q1) **A** Create a DataFrame from a dictionary containing book title, author, category and rating for at least 8 books. Display only the first 5 records using a suitable Pandas function. [10 marks]

Q2) **B** Write a program to create Dataset Name: StudentData.csv [20 marks]

Save the following data in Excel and store it as a .CSV file.

Name	Address	Age	Class
Shyama	Pune	20	BSc (Chemistry)
Shreeji	Mumbai	19	BSc (Physics)
Ramya	Delhi		BSc (Biology)
Radha	Mumbai	21	BSc (Zoology)
Satya	Pune	22	BSc (CS)

Import Dataset

Replace missing values of:

- Age column → with mean age
- Address→with One hot encoding

OR

Q2) **B** Write a program to implement create a DataFrame containing the salaries (in thousands) of 15 employees in a company. Write a Python program to calculate the quartiles (Q1, Q2, Q3) and selected percentiles (25th, 50th, 75th, 90th), and detect outliers using both the IQR and Z-score methods. [20 marks]

Q3) Viva [05 marks]

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Q1) **A** Write a Python program to find the frequency of the three species of the Iris data (Use iris.csv) [10 marks]

Q2) **B** Generate a random array of 50 integers and display them using a line chart, scatter plot, histogram and box plot. Apply appropriate color, labels and styling options. [20 marks]

OR

Q2) **B** Write a Python program to compute complete descriptive statistics for the numeric columns (Salary) using pandas, including mean, median, variance, standard deviation, minimum, maximum, quartiles, and count. [Use employee.csv] [20 marks]

Q3) Viva [05 marks]

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Q1) **A** Create a student DataFrame which contains student_id, name and marks. Write a Python program to calculate mean, median, and mode of given data. [10 marks]

Q2) **B** Write a Python program [20 marks]

1. To create a dataframe containing columns emp_name, age and salary. Add 10 rows to the dataframe.
2. Apply shape (), describe () on given frame
3. To Add 5 rows with duplicate values and missing values. Add a column 'Department' with empty values. Display the data.

OR

Q2) **B** Write a python program to apply discretization on the continuous Marks attribute into categorical performance levels such as *Poor, Average, Good, and Excellent*. Display the dataset before and after discretization. [Use Student_Marks.csv] [20 marks]

Q3) Viva [05 marks]

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Q1) **A** Write a Python program to perform basic data exploration by displaying the shape, size, data types, statistical details, number of observations, and missing values in a dataset. [10 marks]

Q2) **B** Create a dataset data.csv having two categorical columns (the country column, and the purchased column). [15] [20 marks]
a. Apply OneHot coding on Country column.
b. Apply Label encoding on purchased column

OR

Q2) **B** Write a Python program to generate 50 random integers and visualize them using a line chart, scatter plot, histogram and box plot with suitable labels and styling. Then add two outliers and display the updated box plot. [20 marks]

Q3) Viva [05 marks]

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Q1) **A** Create a dictionary that stores student names as values for the key **'Name'**, marks as values for the key **'Marks'**, and city names as values for the key **'City'**. Convert the dictionary into a DataFrame and display it. [10 marks]

Q2) **B** Write a Python program to compute complete descriptive statistics for the numeric columns (Salary) using pandas including mean, median, variance, standard deviation, minimum, maximum, quartiles and count. [Use employee.csv] [20 marks]

OR

Q2) **B** Import dataset "iris.csv". Write a Python program to create a Bar plot to get the frequency of the three species of the Iris data. Write a Python program to create a histogram of the three species of the Iris data. [20 marks]

Q3) Viva [05 marks]

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Q1) **A** Write a python program to create two lists, one representing subject names and the other representing marks obtained in those subjects. Display the data in a pie chart and bar chart. [10 marks]

Q2) **B** Write a Python program to read a CSV file named data.csv and perform following operations [20 marks]

- Display DataFrame
- describe ()
- head ()
- tail ()
- max ()
- min ()

OR

Q2) **B** Write a Python program for Handling Missing Value. Replace missing value of salary, age column with mean of that column. Generate a line plot of name Vs salary [Use Data.csv file]. [20 marks]

Q3) Viva [05 marks]

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Q1) **A** Create a NumPy array containing the ages of 10 employees. Write a Python program to calculate and display the minimum age, maximum age and average age. [10 marks]

Q2) **B** Download any dataset from the UCI repository. Read the CSV file using read_csv() function. Describe the dataset using an appropriate function. Display the mean value of numeric attributes and check whether any data values are missing. [20 marks]

OR

Q2) **B** Create an Employee DataFrame containing Employee_ID, Department, Age and Salary. Implement Simple Random Sampling by selecting 5 employees, Systematic Sampling by selecting every 3rd employee, and Stratified Sampling by grouping employees according to Department and selecting one employee from each department. [20 marks]

Q3) Viva [05 marks]

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Q1) **A** Create two lists representing subject names and marks obtained. Convert the lists into a DataFrame and display the DataFrame. [10 marks]

Q2) **B** Write a Python program to read a CSV file named employee.csv and perform following operations [20 marks]

- Display DataFrame
- describe ()
- head ()
- tail ()
- max ()
- min()

OR

Q2) **B** Create a Matplotlib Figure and Axes object for a given numerical dataset. Set the figure size and customize the title, x-label, y-label, line width and other suitable properties. [20 marks]

Q3) Viva [05 marks]

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Q1)	**A**	Create a DataFrame containing product names, quantities and prices. Display the DataFrame and calculate the total quantity and average price.	[10 marks]
Q2)	**B**	Create a DataFrame containing product prices and perform equal-width binning to divide the prices into three categories: Low, Medium and High. Display the original and binned data.	[20 marks]

OR

Q2)	**B**	Create a numerical dataset and perform unit-length normalization on the data. Display the original data and the normalized values.	[20 marks]
Q3)		Viva	[05 marks]

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Q1)	**A**	Create a NumPy array containing numerical values and standardize it using preprocessing.scale(). Display the original and standardized values.	[10 marks]
Q2)	**B**	Using iris.csv, create a box plot to show how Sepal Length, Sepal Width, Petal Length and Petal Width are distributed across the three species.	[20 marks]

OR

Q2)	**B**	Using iris.csv, create a Seaborn pair plot and analyse the relationships among the different features. Identify which flower species appears to be the most separable.	[20 marks]
Q3)		Viva	[05 marks]